IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2014-15

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected Copy

BIOMEDICAL ELECTRONICS

Friday, 12 December 9:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer 4 out of 5 questions

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : T. Constandinou

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1. This question relates to the design of an instrument for measuring bio-potentials.

a) Sketch a typical ECG waveform labeling each segment. Explain briefly what each segment of the ECG waveform represents during a single heartbeat. [4]

b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record the ECG from the heart. The front end will use the instrumentation amplifier shown in Figure 1.1

(i) Derive an expression for the output voltage V_{out} of the amplifier as a function of the differential voltage V_1-V_2 and select suitable resistor values to give you a gain of -11. You may use R_2=100KΩ. [4]

(ii) Design a suitable band pass filter to extract the signal of interest. You may assume a 60dB/decade roll off for the low-pass and a 20dB/decade roll off for the high pass sections. State equations for cut off frequencies and select suitable resistors. You may assume you have 10nF capacitors to your disposal. [4]

(iii) The instrumentation system will work off a 6 V coin cell battery. Using the circuits of parts b.i, and ii and any additional circuits, sketch the complete schematic of the instrumentation system that utilises the whole dynamic range of the system. Annotate on your schematic values of resistors and capacitors with a justification of the final choices. You may assume V_{DD}=3V and GND=0V and V_{SS}=-3V. [4]

c) i) Show how the circuit in Figure 1.1 can be modified with a "*driven right leg circuit*" to reduce the common mode voltage created by current capacitively coupled to the body from the power-lines. [2]

ii) Show that the common mode voltage with this circuit reduces to:

$$V_{CM} = \frac{i_D R_0}{1+2\frac{R_2}{R_1}}$$

[2]

Table 1.1

Signal	Frequency range (Hz)	Amplitude range (mV)
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

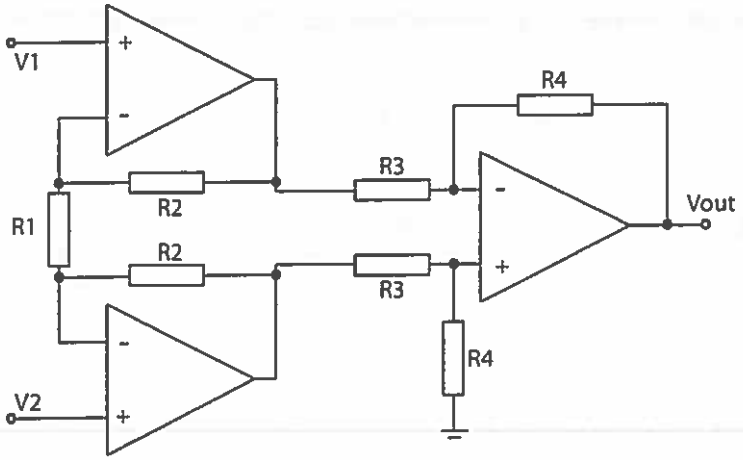


Figure 1.1

2. This question relates to the design of an electrical stimulation circuit for a peripheral nerve interface for upper-limb amputees.

a) Briefly describe and compare *Cuff, TIME,* and *LIFE* electrodes that are applicable to peripheral nervous system (PNS) interfacing. [3]

b) You are required to design a cuff electrode in a multipolar configuration (i.e. with all the electrodes being identical). Given the parameters below, and that only 2 electrodes will be active at any moment in time (i.e. bipolar operation), determine the electrode geometry and sketch your electrode design annotating dimensions. [3]

Nerve diameter = 0.8mm
Independent channels = 3
Electrode material = Pt
Charge density of Pt = 100μC/cm^2/cycle
Maximum stimulus magnitude = 250nC/cycle

c) The desired stimulation pattern is shown in Fig. 2.1. You are required to design a circuit to generate and deliver this stimulus. Assume the total impedance between any pair of electrodes is 10kΩ, and that this impedance is purely resistive.

i. Identify the key blocks and select circuit topologies (and required parameters) to meet the specifications. [3]
ii. Sketch your design (i.e. circuit schematic) [5]
iii. Provide a truth table, state transition diagram and/or combinational logic expressions for the controller. [2]

d) If your stimulator design achieves a charge balance of ~1%, describe how any residual charge can be removed and how this can be incorporated within your design. [2]

e) Discuss how the multipolar electrode scheme can be exploited to improve selectivity (i.e. spatial resolution). [2]

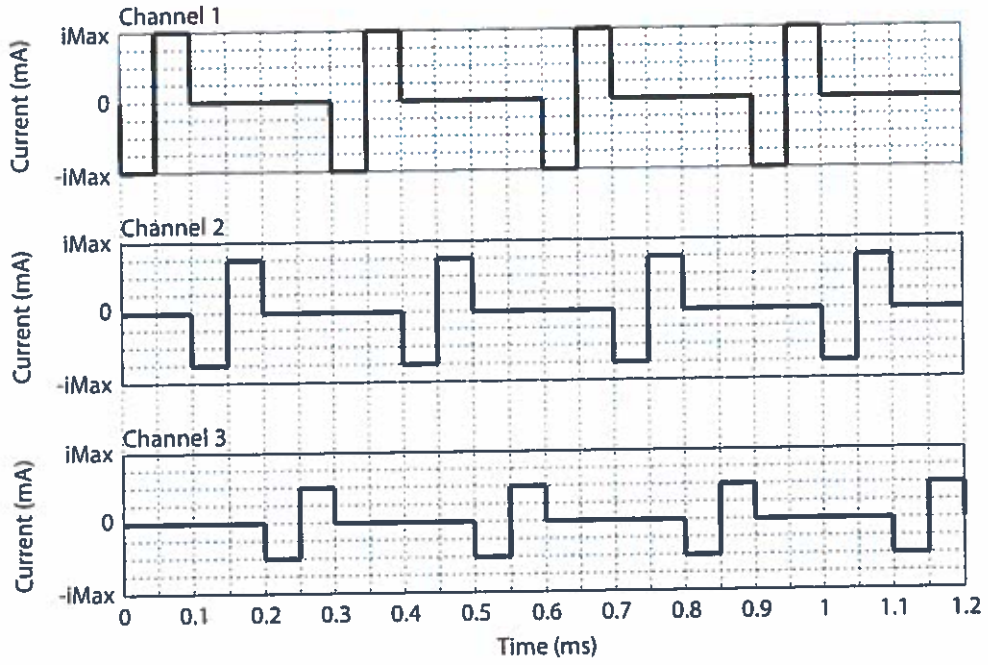


Figure 2.1

3. This question relates to the design of a novel integrated sensor for chemical measurement.

a) Electrochemical measurement has as a basis a redox reaction. Explain what is meant by a redox reaction using reaction equations where necessary. [4]

b) Figure 3.1 shows a new pixel architecture to be used as part of an array in an integrated CMOS based diagnostic device. The total pixel size is 150μm x150μm with a total sensing area of 100μm x 100μm which is capable of sensing both glucose and pH. The working electrode is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7V according to the following reaction:

$$\text{glucose} + O_2 \xrightarrow{\text{glusoce oxidase}} \text{gluconic acid} + H_2O_2 \quad (1)$$

$$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$$

i) The sensor shall be used to measure a physiological glucose range of 0-20mM. The peak redox current for a scan rate of 1V/s is given by the following equation:

$$i_p = 2.69 \times 10^5 n^{3/2} A D^{\frac{1}{2}} C$$

where n is the total number of electrons given in equation 1, the diffusion coefficient D=4.38X10^{-8} m^2/s and A is the total area of the working electrode, WE, in Figure 3.1. Calculate the maximum range of the WE current, using the dimensions shown in Figure 3.1, to capture the physiological range of glucose. [2]

ii) The amperometric sensor in the pixel shall be driven by a potentiostat and transimpedance amplifier which uses an integration capacitor, C_E to convert the sensed current to a voltage. The circuit operates of a single supply with V_{DD}=3.3V and V_{SS}=0V. Sketch the schematic of a suitable measurement system used to interface to the amperometric sensor and label the voltages of all electrodes. [6]

iii) The transimpedance amplifier has a poly-silicon integration capacitor determined by the remainder of the area in the pixel shown in Figure 3.1. A capacitance per unit area of poly-silicon is 4x10^{-4} pF/um^2. Calculate a suitable clock frequency that maximizes the dynamic range of the pixel for glucose measurement. [4]

iv) Sketch a suitable schematic to interface to the pH sensor including switches to interface to more than one pixel. [4]

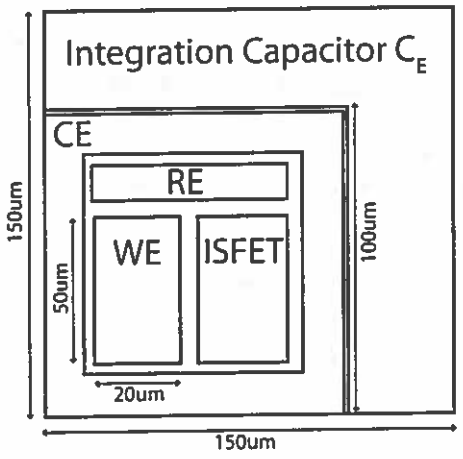


Figure 3.1

4. This question concerns energy harvesting and wireless power supplies for body worn and implanted medical electronics.

a) It is proposed that a medical implant in the abdomen is to be powered from an inertial energy harvester device.

i) Explain how such a device works. Use a diagram and include in your answer a brief discussion of the possible transducer technologies that could be used.

[4]

ii) The power output of a harvester at resonance can be shown to be:

$$P_{res} = \frac{1}{2} Y_0 \omega^3 m Z_l$$

Where Y_0 is the amplitude of the driving motion, ω is the driving frequency, m is the proof mass and Z_l is the amplitude of the mass travel. Explain why it is logical that the power is proportional to:

- the proof mass
- the cube of frequency

[3]

iii) The position profile at the place of the proposed implant was measured as shown in Figure 4.1. Estimate the output power of a resonant harvester attached at this location with the a proof mass of 1 g and a maximum amplitude of mass displacement of 1 mm. State the assumptions you are making.

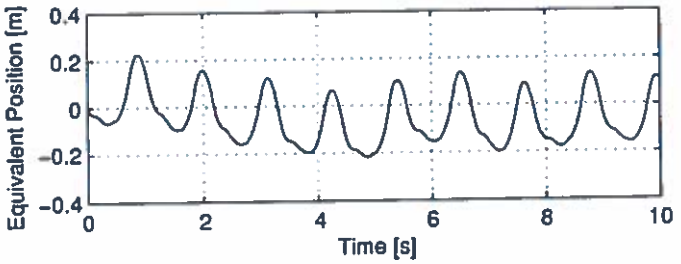


Figure 4.1 Position of generator attachment point whilst walking

[4]

iv) In reality the power output will be less than this. State two effects that reduce the power output of the system.

[2]

b) A wireless power transfer system is to be used to power an implanted actuator which closes the valve between the stomach and oesophagus in a person with acid reflux disease.

i) A suggestion is made to increase the link efficiency of the system by adding ferromagnetic materials to the coils. Explain why this may increase efficiency.

[3]

ii) It is then proposed that the Q-factors can be increased by operating the system at higher frequency. Explain why this is thought, and what factors should be considered when increasing the system frequency.

[4]

5. This question relates to the design of an actuated prosthetic hand with sensory feedback for amputees.

a) Describe the three different categories (i.e. sensory, cognitive and motor) of neural prosthesis, identifying the similarities or differences between these. Give one example for each. [6]

b) A hand prosthesis would combine a sensory and motor prosthesis. Sketch a block diagram of the sensory part of the prosthetic device only, given this will interface with the peripheral nervous system. [6]

c) Given that this prosthesis will utilize a new type of tactile sensor (i.e. to sense touch) on each fingertip, describe how the sensory signal path may be mapped from sensor to peripheral nerve given the following dynamics. [4]

Tactile sensor range: 1mN to 25N
Interface circuit dynamic range: 40dB (sensor input range is electronically configurable)
Current-mode stimulation range: 0.1mA to 1.2mA

d) Given that the peripheral nerve interface will require an implantable battery, select the most suitable option, given the following characteristics. Justify your choice and state your assumptions (hint: you should estimate the duty cycle based on an average daily usage). [4]

Required power supply (for implanted electronics): 7V
Standby power consumption: 100uA
Additional power consumption when sensing (i.e. and also stimulating): 2.5mA

Option 1: Rechargeable battery: voltage=3V, capacity = 50mAh, total recharge cycles = 1000
Option 2: Non rechargeable battery: voltage=3.7V, capacity = 10Ah